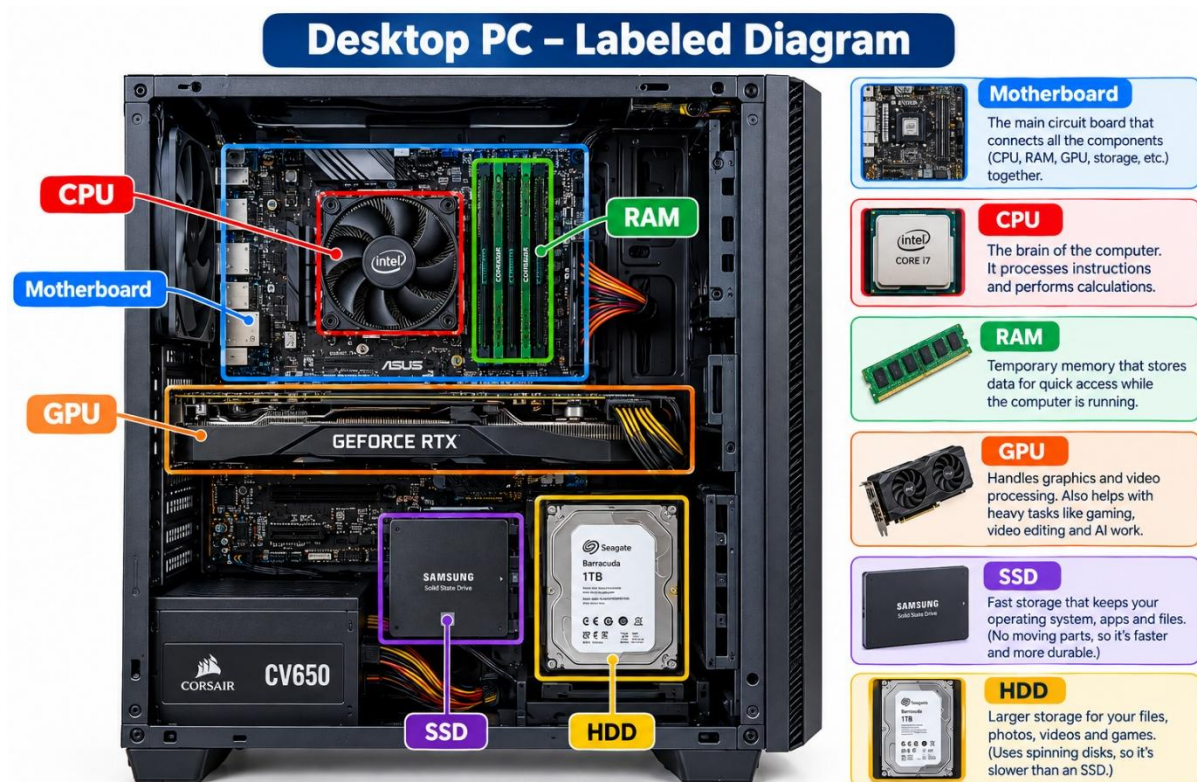


Day 1 : Overview of Hardware ComponentsTheory

Task 1 : Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



Task 2 : Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Full Form	Hard Disk Drive	Solid State Drive
How it works	Uses spinning disks to store data	Uses memory chips to store data
Speed	Slower	Much faster
Boot Time	Computer starts more slowly	Computer starts quickly
File Transfer	Slower	Faster
Durability	Less durable because it has moving parts	More durable because it has no moving parts
Shock Resistance	Can be damaged by drops or strong shocks	More resistant to shocks and drops
Noise	Can make some noise while working	Silent
Power Use	Uses more power	Uses less power
Heat	Produces more heat	Produces less heat
Price	Cheaper	More expensive
Storage Capacity	Large capacity is available at lower cost	Large capacity costs more
Common in	Desktop PCs, external storage, backups	Laptops, gaming PCs, modern desktops
Gaming	Games load more slowly	Games load faster
Best Use	Storing many files at low cost	Fast performance and quick loading

Task 3 : Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

1. CPU (Central Processing Unit)

The CPU is the brain of the computer. In games like PUBG or Free Fire, it handles important tasks such as player movement, game rules, enemy actions, physics, and other calculations. A powerful CPU helps the game respond quickly and run smoothly.

2. RAM (Random Access Memory)

RAM is the temporary working memory of the computer. It stores the game data that is currently being used, so the CPU can access it quickly. More RAM can help the game run smoothly, especially when other applications are running in the background.

3. GPU (Graphics Processing Unit)

The GPU is responsible for the game's graphics. It creates the images you see on the screen, including characters, buildings, weapons, shadows, lighting, and effects. A better GPU can give you better graphics quality and higher FPS, making the game look smoother.

Easy Way to Remember

CPU = Thinks and Calculates 🧠

RAM = Holds Working Data 📦

GPU = Creates Graphics 🎮

Task 4 : Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Hardware Upgrades for Running FIFA Smoothly

1. GPU (Graphics Card):

I would upgrade the GPU because it handles the game's graphics, such as players, stadiums, grass, lighting, shadows, and effects. A better GPU helps provide higher FPS and smoother gameplay.

2. CPU (Processor):

I would upgrade the CPU because it handles game calculations, player movements, game logic, and other important tasks. A powerful CPU helps the game run smoothly and respond quickly.

3. RAM:

I would increase the RAM because FIFA needs enough memory to run properly. 16 GB RAM is a good choice for smooth gaming and also allows other programs to run in the background.

4. SSD:

I would use an SSD instead of an HDD because an SSD is much faster. It helps FIFA and Windows start faster and reduces loading times.

5. PSU (Power Supply Unit):

I would check or upgrade the PSU if I install a more powerful GPU or CPU. A suitable PSU provides enough power for all the components and helps the PC work safely.

6. Cooling System:

I would also check the cooling system because gaming can make the CPU and GPU hot. Good cooling helps prevent overheating and keeps the PC performing well during long gaming sessions.

Upgrade Priority

- 1. GPU** – Better graphics and FPS
- 2. CPU** – Better game calculations and stable performance
- 3. RAM** – Better multitasking and smoothness
- 4. SSD** – Faster loading and overall responsiveness
- 5. PSU** – Provides safe and sufficient power
- 6. Cooling** – Helps prevent overheating